

FEASIBILITY AND VIABILITY

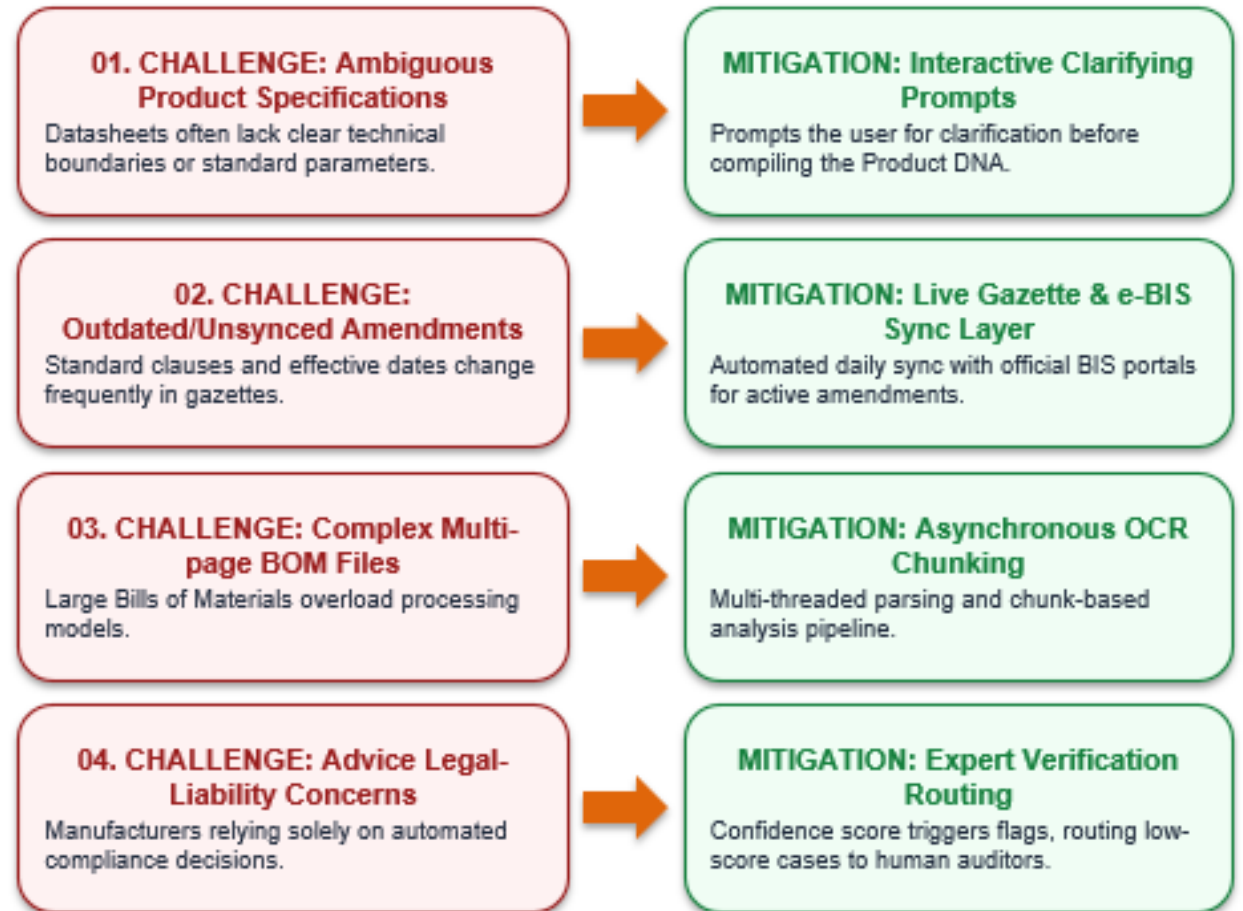
• Feasibility Analysis

- Technical: RAG, vector DBs & fine-tuned LLMs are production-ready.
- Data: BIS publishes standards in PDF; OCR + layout parsing extracts clauses.
- Market: 35,000+ BIS licensees & millions of SMEs need compliance guidance.
- Scalability: Cloud-native architecture scales from single-user to enterprise API.

• Viability

- Revenue: Freemium SaaS + Enterprise API + Lab partnership fees.
- Growth: Expand from BIS to ISO, IEC, ASTM & other standard bodies.
- Adoption: Direct integration with BIS portal & MSME ministry platforms.
- Sustainability: Subscription model ensures continuous update coverage.

Real-World Challenges & Mitigation Strategies



IDEA TITLE

BIS Compliance Compiler

AI-first compliance assistant translating product artifacts into clause-level BIS applicability, tests, evidence & lab actions.

Input Sources:

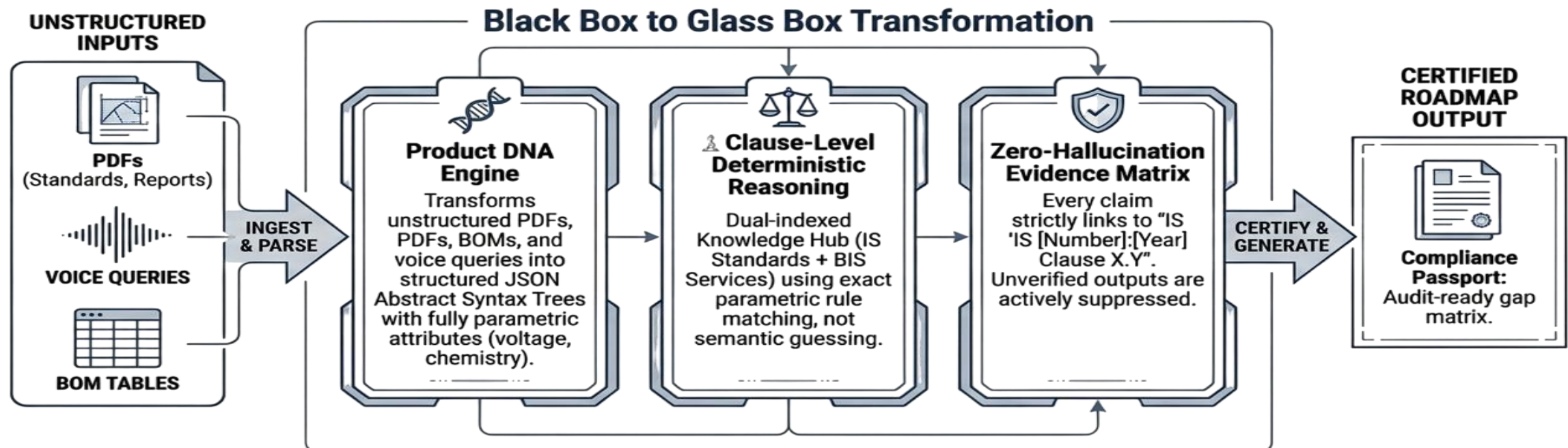
Text | Voice | PDF | BOM |
Images (OCR)

Core Engine:

Product DNA Engine + BIS Knowledge
Engine (clause-level logic)

Outputs:

Compliance Passport | Gap Report |
Evidence Graph | Lab Recommendations



Technologies Used

01

FRONTEND & GRAPH

React.js
Tailwind CSS
React Flow

02

BACKEND ENGINE

Python 3.12
FastAPI
Pydantic

03

DATABASE & KB

PostgreSQL
pgvector
SQL Stores

04

INGESTION & PARSING

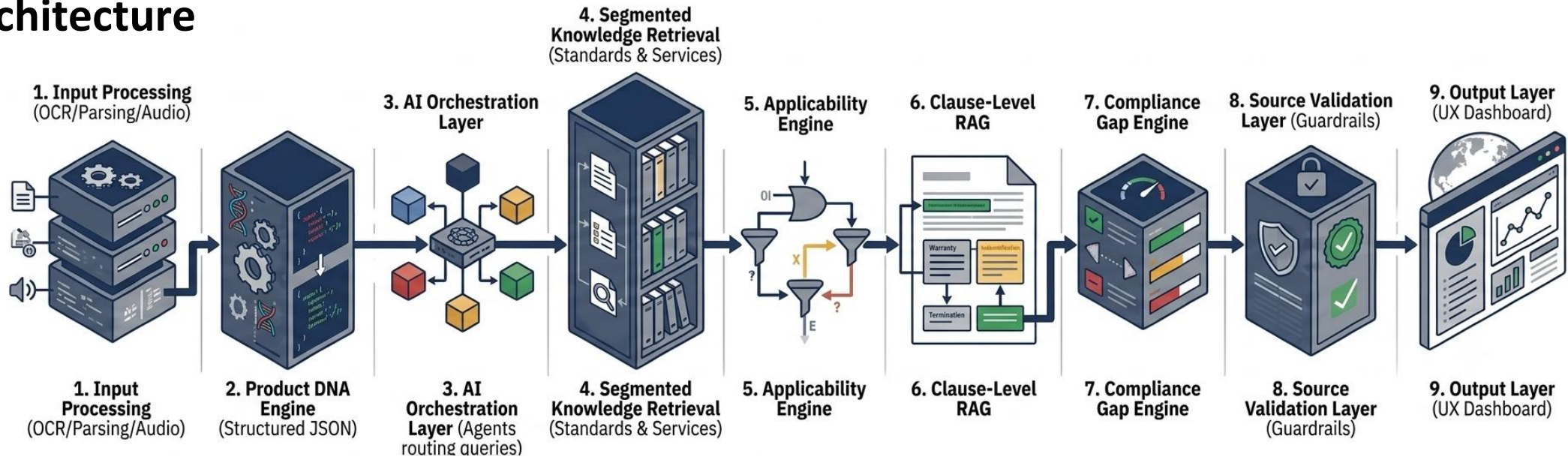
PyMuPDF
Tesseract OCR
Whisper STT

05

REASONING AI

Structured LLM
Instructor
Citation Guard

Architecture



TIME REDUCTION

95%

3 Weeks of Gazette
Reading & Consulting

3 Minutes of
Deterministic Analysis

✂ **Instant Compliance**

COST SAVINGS

80%

₹50K - ₹5L standard
consultant fees per
product registration

₹0 - ₹10K maximum
Self-Compiled audit cost

📉 **Slashes MSME Barrier**

MSME BENEFICIARIES

63M+

Democratizes access
for small manufacturers,
startups, and testing labs
across all 20,000+
Indian Standard sectors.

🏠 **Grassroots Growth**

MARKET OPPORTUNITY

₹31.5T

Unlocks dormant local
manufacturing capability
by speeding up standard
adoption & reducing local
go-to-market barriers.

🌐 **Trillion-Dollar Impact**

ECONOMIC IMPACT

Unlocking Domestic Value Chains

- Speeds time-to-market by 4 to 8 weeks
- Direct capital redistribution back to MSME research
- Reduces operational overhead for standard audits
- Boosts global competitiveness of local products

SOCIAL IMPACT

Democratizing Regulatory Access

- Vernacular support bridges language gap for rural tech
- Levels playing field between startups & corporate giants
- Simplifies technical engineering jargon into plain logic
- Fosters self-reliance (Atmanirbhar Bharat) at grassroots

GOVERNANCE & TRUST

Deterministic Zero-Trust Accuracy

- 100% auditable citation links (Clause-Level Provenance)
- Slashes counterfeit/non-conforming manufacturing
- Integrates cleanly into Manakonline & e-BIS portals
- Lowers testing friction with automated lab allocation



References

- Bureau of Indian Standards (BIS) - <https://www.bis.gov.in>
- Manakonline / e-BIS certification portal
- BIS Act, 2016 & BIS Rules, 2018
- IS 15111 (2002) – General requirements for product certification
- Indian Standards (IS) database & Quality Control Orders (QCOs), Ministry of Consumer Affairs
- Lewis, P., et al. (2020). ‘Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.’ NeurIPS 2020.
- Gao, L., et al. (2023). ‘RAG vs. Fine-tuning: Pipelines for Domain-Specific QA.’ arXiv:2312.05934.
- OpenAI (2024). ‘Best Practices for Retrieval-Augmented Generation Systems.’
- Ministry of MSME, Govt. of India – Annual Report 2024-25
- NITI Aayog (2023). ‘Responsible AI for All’ – Guidelines for trustworthy AI in India
- Open-source documentation: FastAPI, PostgreSQL/pgvector, Whisper STT

Comparison with Existing Solutions

Feature	BIS Compliance Compiler	Manual Consultants	Generic LLM Tools	BIS Portal
Clause-Level Retrieval	✓	✗	✗	✗
AI-Powered Gap Detection	✓	✗	✗	✗
Evidence Validation	✓	✗	✗	✗
Hallucination Guard	✓	✗	✗	✗
Multi-Modal Input (Voice/PDF)	✓	✗	✓	✗
Automated Lab Recommendations	✓	✗	✗	✗
Real-Time Standard Updates	✓	✗	✗	✗